

EE45: Intro to Foundations of ML

Mod 1 - Lec 1: Data Fitting Concepts

Course Overview Reminder

- Focus on
 - ▶ **simple, core models**: e.g., linear regression & classification),
 - ▶ **fit models via optimization**: gives us a framework for learning from data, generalization, and interpretability
 - ▶ and the **mathematics of ML**: enables us to understand (transparently) and improve models

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 - ▶ and the **mathematics of ML**: enables us to understand (transparently) and improve models
- Show connections to applications in
 - ▶ data science & statistics,
 - ▶ signal processing,
 - ▶ decision-making & control,
 - ▶ computer vision/perception,
 - ▶ AI and robotics

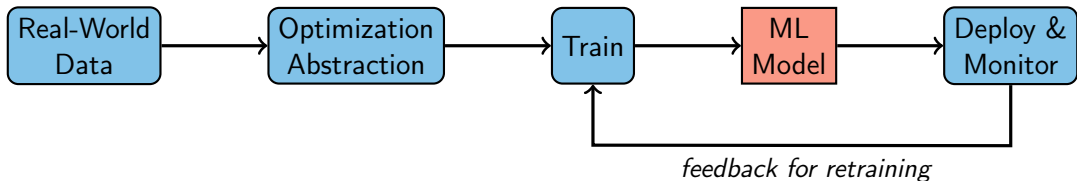
Overview: [Module 1] Data Fitting Concepts

- This week
 - ▶ How to measure "fit" of a model to data?: norms, distances, inner products
 - ▶ Example: k -means clustering

References:

- [VMLS]: Chapters 1–3
- Topics: Vectors, Inner product, Linear functions, Regression model, Norm and distance

ML workflow: Role of Linear Algebra & Optimization



This course: Show that many interesting learning and optimization problems are framed and solved using language and techniques from **Linear Algebra** and **Optimization**

ML workflow: Role of Linear Algebra & Optimization

Real world task involving data $\longrightarrow$ Optimization Problem $\longrightarrow$ ML model

Canonical Optimization Problem Terminology:

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-
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Formulating a Learning Problem as Optimization

- The quality of a model is measured by how well it fits the data.

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- Machines need to measure similarity, and distances are the key
 - ▶ Norms and distances turn subjective ideas (are these two cats alike?) into math: $\|x - y\|$.
 - ▶ Different norms capture different ideas of similarity (Euclidean vs Manhattan vs cosine).

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Model

A model is a recipe or rule that takes input data and produces an output (a prediction).

Model fitting as optimization

Goal

Given data $\{(x_i, y_i)\}_{i=1}^n$ and a model f_θ , choose parameters θ to best explain the data.

Key idea: *measuring error* $\longleftrightarrow$ *optimization objective*

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Then we **optimize** that number:

$$\theta^* = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_{\theta}(x_i))$$

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Design choice: the loss ℓ is usually built from a **distance** (how far?) or a **similarity** (how aligned?).

Formulating Optimization: Similarity & "Distance" Measures

- Similarity is the hidden engine behind search, recommendation, and personalization.
 - ▶ Netflix recommends shows based on similarity to things you've watched.
 - ▶ Google finds web pages since they're similar in content to your query.
 - ▶ Spotify suggests songs since your playlist is similar to other users' playlists.

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- Fundamental ML Operations Depend on It
 - ▶ **Classification:** Assign a label to a new point based on similar past examples (nearest neighbors).
 - ▶ **Clustering:** Group data points that are similar (k-means).
 - ▶ **Dimensionality reduction:** Keep points that are similar in the original space close together in a compressed representation (PCA, t-SNE).
 - ▶ **Representation learning (modern ML):** Train embeddings so that similar meanings (words, images, faces) are close in vector space.

Similarity

- How do we decide if two people, two images, or two signals are similar?
- examples: Two histories of Spotify users', two patients' health metrics, etc.
 - ▶ Two grayscale digit images of "3" and "8".



- represent each object as a vector in some space. Similarity is then about distance between vectors.

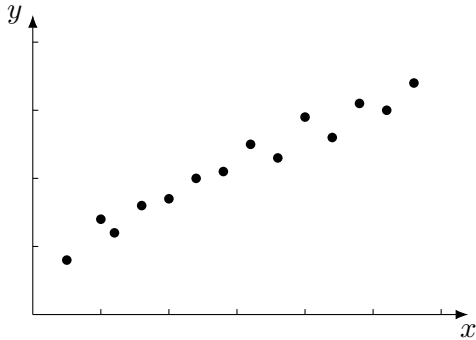
Digit 3 Vector Strip



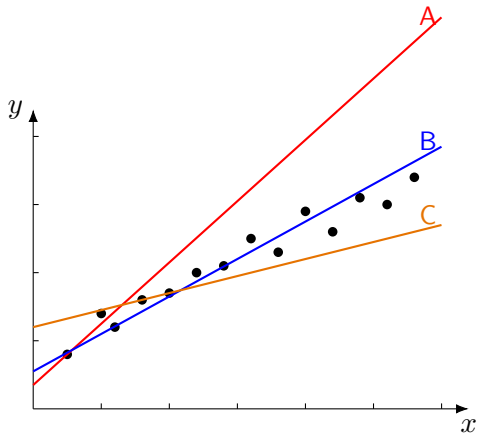
Digit 8 Vector Strip



Model "Fit": Which line fits these points best?

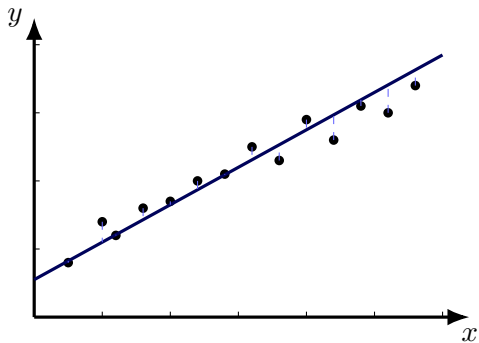


Model "Fit": Which line fits these points best?



Which line would you pick, and why?
How can we make this decision mathematically?

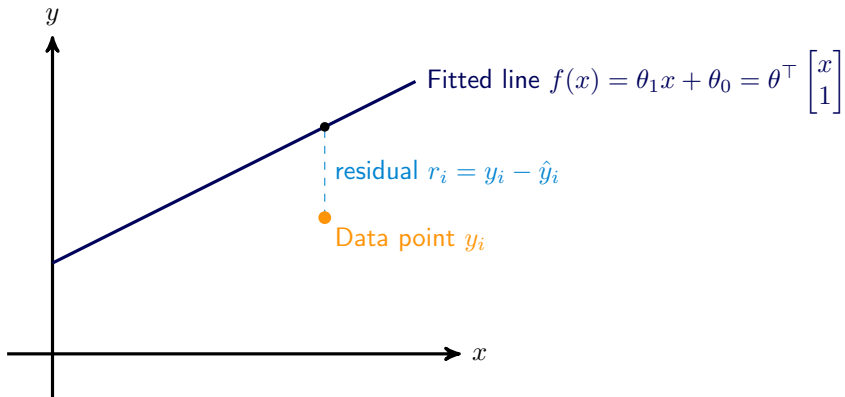
Residuals: distance from points to a line



Residuals (vertical distances) to candidate B.
Least squares picks the line that minimizes the sum of squared residuals.

$$r_i = y_i - \hat{y}_i$$

Residuals



Least Squares Vector Notation

This problem is easily written in vector notation.

where

-
-
-

Vectors: notation

- a vector is an ordered list of numbers: $(-1.2, 0, 3.6)$, or

$$x = \begin{bmatrix} -1.2 \\ 0 \\ 3.6 \end{bmatrix}$$

Examples

- (x_1, x_2) represents a location or a displacement in 2D
- examples:
 - ▶ portfolio: entries give shares (or \$ value or fraction) held in each of n assets, with negative meaning short positions
 - ▶ cash flow: x_i is payment in period i to us
 - ▶ sampled audio signal: x_i is the acoustic pressure at sample time i
 - ▶ feature vector: x_i is the value of i th feature or attribute of an entity
 - ▶ word count vector

Word count example

- a short document:
“Word count vectors are used in computer based document analysis. Each entry of the word count vector is the number of times the associated dictionary word appears in the document.”
- dictionary & word count vector:
word
in
number
flower
the
document

Length of a vector

- We now have a way of writing data as vectors.
- We now need a way of measuring the *length* of a vector, and then distances between vectors.
- The typical length of a vector x is given by the Euclidean norm:

Norm and distance

- A norm is a function that measures the length of a vector.
- Euclidean norm (2-norm) is $\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$
- properties: $\alpha \in \mathbb{R}$ is a scalar, $x, y \in \mathbb{R}^n$ are vectors
 - ▶ homogeneity:
 - ▶ triangle inequality:
 - ▶ nonnegativity:
 - ▶ definiteness:

Inner Products

- inner product (or dot product) $\langle x, y \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ of n -vectors x and y is

$$\langle x, y \rangle = x^\top y = \sum_{i=1}^n x_i y_i$$

satisfying:

- ▶ **Linearity:**
 - ▶ **Symmetry:**
 - ▶ **Positivity:**
- Examples:

Linear combinations

- For vectors $a_1, \dots, a_n$, scalars $\beta_1, \dots, \beta_n$,

$$\beta_1 a_1 + \dots + \beta_n a_n = \sum_{i=1}^n \beta_i a_i \quad \beta_i \text{ are weights, sum is a "mixture" of the } a_i$$

is a linear combination with coeff's $\beta_1, \dots, \beta_n$.

- 2D example: $b = 0.75a_1 - a_2$
- Line and segment: a, b are n -vectors,
 $c = (1 - \theta)a + \theta b$
 - ▶ any $\theta \in \mathbb{R} \implies c$ on line through a, b
 - ▶ $0 \leq \theta \leq 1 \implies c$ on line segment ab

Matrix–Vector Multiplication as Inner Products

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} - & a_1^\top & - \\ - & a_2^\top & - \\ - & a_3^\top & - \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} a_1^\top x_1 \\ a_2^\top x_2 \\ a_3^\top x_3 \end{bmatrix} = \begin{bmatrix} \langle a_1, x \rangle \\ \langle a_2, x \rangle \\ \langle a_3, x \rangle \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

where

$$y_i = \langle a_i, x \rangle = \sum_{j=1}^3 a_{ij} x_j$$

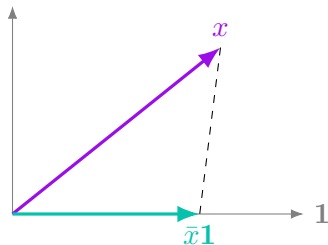
Standardizing Data

Example: Mean as Inner Product

- The mean is the “balance point” (center of mass).
- The sample mean is

$$\mu = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{\langle x, \mathbf{1} \rangle}{\langle \mathbf{1}, \mathbf{1} \rangle}$$

- Geometrically: $\bar{x} \cdot \mathbf{1}$ is the **projection** of x onto the "span" of $\mathbf{1}$
- Intuition: the mean is the "best constant approximation" of the data

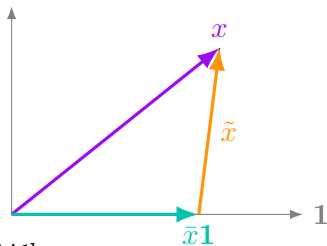


Centering Feature Vectors

Given $x \in \mathbb{R}^n$ and $\mathbf{1} = (1, 1, \dots, 1)^\top \in \mathbb{R}^n$:

$$\tilde{x} = x - \bar{x} \mathbf{1} = x - \frac{1}{n} \mathbf{1}^\top x \mathbf{1} = x - \frac{1}{n} \mathbf{1} \mathbf{1}^\top x = \left(I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top \right) x$$

- $\mathbf{1} \mathbf{1}^\top \in \mathbb{R}^{n \times n}$ is the all-ones matrix.
- $H = I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top$ is the **centering matrix**.
- H is symmetric and idempotent ($H^2 = H$): an **orthogonal projection**.
- Geometrically: projects x onto the subspace orthogonal to $\mathbf{1}$.



Example: Variance and Standard Deviation

- The centered data vector is $x - \mu \mathbf{1}$
- The standard deviation is the “typical distance” of points from that balance point.
 - ▶ Small SD $\rightarrow$ data clustered tightly around the mean.
 - ▶ Large SD $\rightarrow$ data spread out.
- If you pick a random data point, the standard deviation is roughly the size of the “typical error” you’d make if you always guessed the mean.
- Think of the mean as a reference position. The standard deviation is like the moment of inertia—how far mass is spread out from the center. Same “center,” but very different spread.
- The variance is the **scaled average squared deviation**

Example: Variance and Standard Deviation

- The centered data vector is $\tilde{x} = x - \bar{x}\mathbf{1} = Hx$, where $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ is the centering matrix.
- Its squared norm gives the total squared deviation:
- The variance is the **scaled average squared deviation**:
- Standard deviation (= avg length of the projected (centered) vector):

Standardizing Data

Given data $x = (x_1, \dots, x_n)$ with mean $\bar{x}$ and std. deviation s , the **standardized data** (z-scores) are

Properties

- Centered: $\text{mean}(z) = 0$
- Scaled: $\text{sd}(z) = 1$
- Interpretable in **standard deviation units**.
- Comparable across different variables

Standardizing Data (z-scores)

Given $x \in \mathbb{R}^n$, mean $\bar{x}$, and centering matrix the **centered data** is

The **standard deviation** is

Standardized data (z-scores):

Why (and When Not) to Standardize Data

- **Why standardize:**
 - ▶ **Comparability across features** Different units (kg, \$, counts) → put in common **std dev units**.
 - ▶ **Geometry matters** kNN, SVM, PCA use distances/dot products; large-scale features dominate otherwise.
 - ▶ **Optimization stability** Gradient descent converges faster; loss surface becomes “rounder.”
 - ▶ **Interpretability** Coefficients are comparable across features.

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- **When **not** to standardize:**

- ▶ **Categorical features** encoded as one-hot or integers → standardizing breaks their meaning.
- ▶ **Features with physical meaning in the model** e.g. age in years, probability values, counts with direct interpretability.
- ▶ **Tree-based models** (decision trees, random forests, gradient boosting) → splits are order-based, not distance-based; scaling irrelevant.

A few more examples

- w is weight vector, f is feature vector; $w^\top f$ is score
- p is vector of asset prices, s gives portfolio holdings (in shares); $p^\top s$ is total portfolio value
- c is cash flow, d is discount vector (with interest rate r):

$$d = \left(1, \frac{1}{(1+r)}, \dots, \frac{1}{(1+r)^{n-1}} \right)$$

$d^\top c$ is net present-value of cash flow

Inner Products Induce Norms

Norms from Inner Products

- Every inner product induces a norm:

$$\|x\| = \sqrt{\langle x, x \rangle}$$

- In the standard inner product:

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

- Interpreted as the **length** of a vector in Euclidean space

Properties of Norms and Distances

- norm of block vectors:

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \|(a, b, c)\|^2 = a^\top a + b^\top b + c^\top c = \|a\|^2 + \|b\|^2 + \|c\|^2$$

where $a \in \mathbb{R}^m$, $b \in \mathbb{R}^n$, $c \in \mathbb{R}^p$.

- (Euclidean) *distance* between vectors a, b is:
- triangle inequality:

Norms

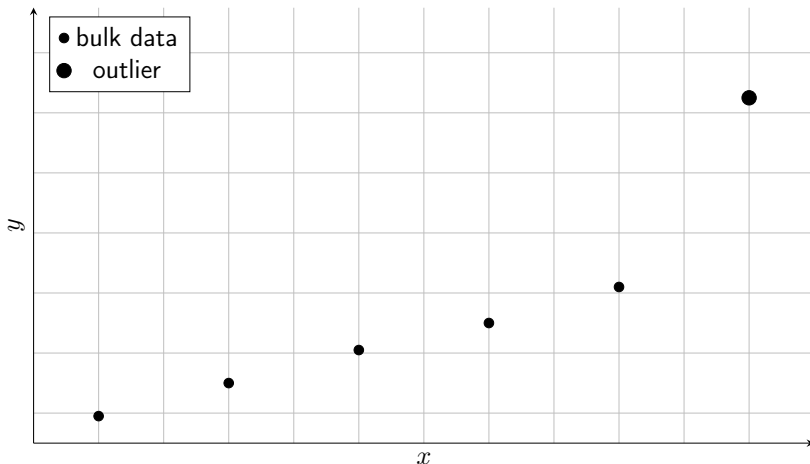
- The Euclidean norm is a special case of a more general class of norms.

Define residual vector $r = (r_1, \dots, r_n)^\top \in \mathbb{R}^n$.

Choosing Norms

- $\|\cdot\|_2$: “mean fit” $\rightarrow$ sensitive to outliers.
- $\|\cdot\|_1$: “median fit” $\rightarrow$ more robust.
- $\|\cdot\|_\infty$: “worst-case fit” $\rightarrow$ minimizes maximum deviation.

Choosing Norm Impacts Model Fit



Note: Without this outlier, ℓ_2 and ℓ_1 give nearly identical fits. The outlier is the stress test.

Norm and distance

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- triangle inequality:

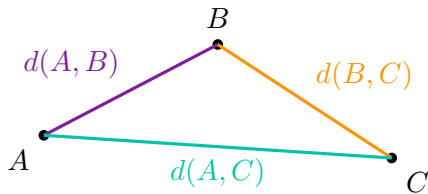
B
•

A
•

•
 C

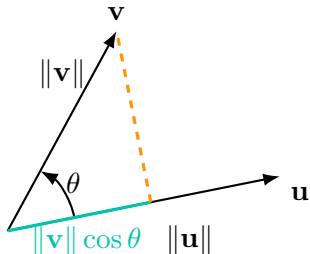
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$$d(A, C) \leq d(A, B) + d(B, C)$$

Cauchy-Schwarz inequality

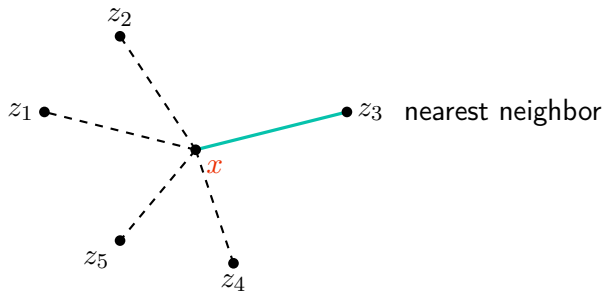


$$|\langle \mathbf{u}, \mathbf{v} \rangle| = \|\mathbf{u}\| \|\mathbf{v}\| |\cos \theta| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

Feature distance and nearest neighbors

- if x, y are feature vectors for two entities, $\|x - y\|$ is the *feature distance*
- for vectors $z_1, \dots, z_m$, z_j is nearest neighbor of x if

$$\|x - z_j\| \leq \|x - z_i\|, \quad i = 1, \dots, m$$



Example: document dissimilarity

- 5 Wikipedia articles: 'Veterans Day', 'Memorial Day', 'Academy Awards', 'Golden Globe Awards', 'Super Bowl'
- word count histograms, dictionary of 4423 words (more setup details in sec 4.4.5 of VMLS)
- pairwise distances shown below:

	Veterans' day	Memorial day	Academy A.	Golden G.	Super Bowl
Veterans' day	0	0.095	0.130	0.153	0.170
Memorial day	0.095	0	0.122	0.147	0.164
Academy A.	0.130	0.122	0	0.108	0.164
Golden G.	0.153	0.147	0.108	0	0.181
Super Bowl	0.170	0.164	0.164	0.181	0

- we'll revisit this example later